

Does the $+$ term diverge?

The z -integral computed, not power-counted

$$a \equiv x' - y', \quad b \equiv y - w, \quad \xi \equiv p^+ / k^+, \quad \text{ph} \equiv e^{-i\xi k \cdot (y' - z)}$$

1. What is now right

The leftover bracket is gone from the second term — that was the one outright error, and it is fixed. Everything else about the split checks out:

- the $+$ sign of the second term (the minus in the bracket against the overall minus);
- the prefactor $\frac{1}{8\pi^5} = \frac{1}{4\pi^4} \cdot \frac{1}{2\pi}$;
- the numerator $(y' - z) \cdot (x - z) (x' - y') \cdot (y - w)$ as the $\delta_{jm} \delta_{ik'}$ contraction;
- the ξ -phase correctly absent from the second term, since $g(0)$ has phase 1;
- the colour structure repeated identically in both terms — correct, it is p^+ -independent;
- **and keeping the lower limit at Λ in the $+$ term makes the split exact**, not just exact to $O(\Lambda)$. Checked: with $\Lambda/k^+ = 10^{-2}, 10^{-4}, 10^{-6}$ the two sides agree to 10^{-31} , whereas pushing the lower limit to 0 costs $1.7 \times 10^{-2}, 1.7 \times 10^{-4}, 1.7 \times 10^{-6}$ respectively. So $\int_{\Lambda}^{\vee - k^+}$ is the right thing to write.

2. The question: does the first term's z -integral converge?

Power counting said no last time. Here is the integral itself. Write the z -dependence of the three index structures,

$$S_1 = \frac{(y' - z) \cdot a (x - z) \cdot b}{(y' - z)^2 (x - z)^2}, \quad S_2 = \frac{(y' - z) \cdot (x - z)}{(y' - z)^2 (x - z)^2}, \quad S_3 = \frac{(y' - z) \cdot b (x - z) \cdot a}{(y' - z)^2 (x - z)^2},$$

and compute the *truncated* integral $F(R) = \int_{|z| < R} d^2 z f(z)$ for each. A convergent integral has $dF/d\log R \rightarrow 0$; a logarithmic divergence has a constant nonzero slope.

In the first term S_1 (on $\frac{1}{p^+ + k^+}$) and S_3 (on $\frac{1}{k^+}$) keep the full phase, because the prescription touches only the middle structure. S_2 (on $[\frac{1}{p^+}]_+$) has its phase replaced by $\text{ph} - 1$.

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=== xi = 0.3 ===
structure          F(1e2)          F(1e3)          F(1e4)
S1 x ph            -11.31311      -20.04642i     -11.33429      -20.05237i     -11.33371      -20.04930i
S2 x ph            +2.87315       -0.11583i      +2.86774       -0.11525i      +2.86783       -0.11568i
S3 x ph            -9.68428       +20.89552i     -9.70475       +20.89741i     -9.70418       +20.90045i
S2 x (ph-1)        -20.13973       -0.11583i     -34.61343      -0.11525i     -49.08091      -0.11568i
S2 bare (= g(0))   +23.01288        +0.00000i      +37.48117      +0.00000i      +51.94874      +0.00000i
dF/dlogR (real)
S1 x ph            dF/dlogR =      -0.00920      +0.00025
S2 x ph            dF/dlogR =      -0.00235      +0.00004
S3 x ph            dF/dlogR =      -0.00889      +0.00025
S2 x (ph-1)        dF/dlogR =      -6.28585      -6.28315
S2 bare (= g(0))   dF/dlogR =      +6.28350      +6.28319
(-2 pi = -6.28319)

=== xi = 0.05 ===
structure          F(1e2)          F(1e3)          F(1e4)
S1 x ph            -39.69317      -7.49804i     -38.63679      -7.50688i     -38.63874      -7.50648i
S2 x ph            +12.84205      -0.08258i     +13.19175      -0.08833i     +13.18988      -0.08831i
S3 x ph            -39.60552      +8.07372i     -38.54912      +8.02315i     -38.55107      +8.02319i
S2 x (ph-1)        -10.17083      -0.08258i     -24.28942      -0.08833i     -38.75886      -0.08831i
S2 bare (= g(0))   +23.01288        +0.00000i      +37.48117      +0.00000i      +51.94874      +0.00000i
dF/dlogR (real)
S1 x ph            dF/dlogR =      +0.45878      -0.00084
S2 x ph            dF/dlogR =      +0.15187      -0.00081
S3 x ph            dF/dlogR =      +0.45879      -0.00085
S2 x (ph-1)        dF/dlogR =      -6.13162      -6.28400
S2 bare (= g(0))   dF/dlogR =      +6.28350      +6.28319
(-2 pi = -6.28319)

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The verdict is unambiguous. S_1 and S_3 with their phases, and S_2 with its phase, all have $dF/d\log R \rightarrow 0$: those z -integrals converge. The subtracted combination $S_2 \times (\text{ph} - 1)$ has $dF/d\log R \rightarrow -2\pi$, the same slope as the bare $S_2 = g(0)$ — because the -1 carries no phase and what is left under $\int d^2 z$ is $\mathcal{K}(x, y'; z)$, the BK real-emission kernel, falling as $1/z^2$.

So **yes: the first term has a transverse divergence** — logarithmic, at large $|z|$, present for every value of p^+ , with coefficient -2π times the spectator factor $\frac{(x'-y') \cdot (y-w)}{(x'-y')^2 (y-w)^2} e^{-ik \cdot (y'-w)}$, which does not vanish on integration over y' .

And the second term carries exactly the same divergence — it is $g(0)$. The two cancel only in the sum. Before the split, the z integral converged for every $\xi > 0$ (the phase cut it at $|z| \sim k^+/(p^+|k|)$) and the whole thing was finite for $\Lambda > 0$, of size $O(\ell_k^2)$. The split has turned one finite object into two infinite ones.

3. So the expression is exact, but not usable

Nothing in it is *wrong* any more — it is an exact rearrangement of a finite quantity. But neither piece exists on its own, so you cannot call the first term “a finite contribution to the cross section” and set it aside, and you cannot absorb the second term into evolution while the first is still infinite.

What has to happen first

1. **Bring in the virtual diagrams of the same order.** They complete the real kernel to the full BK kernel,

$$\mathcal{M}(x, y'; z) = \frac{(x - y')^2}{(x - z)^2 (y' - z)^2} = \frac{1}{(x - z)^2} + \frac{1}{(y' - z)^2} - 2\mathcal{K}(x, y'; z),$$

whose large- ρ angular average cancels at order ρ^{-2} ($1 + 1 - 2 = 0$) so that it falls as ρ^{-4} . Then $g(0)$ has a convergent z integral and both pieces of the split are finite separately. **This is the only real fix** — no choice of prescription can substitute for it.

2. **Then** subtract with $\theta(k^+ - p^+)$ rather than over the whole range,

$$\left[\frac{1}{p^+} \right]_+^{(k^+)} g = \frac{g(p^+) - \theta(k^+ - p^+)g(0)}{p^+},$$

which isolates $\ell_k = \log \frac{k^+}{\Lambda}$ instead of $\log \frac{\vee - k^+}{\Lambda}$ and gives each piece a finite $\vee \rightarrow \infty$ limit. (As written, both of your terms slide with $\vee$ by $\mp \log \vee$, even though their sum does not.)

The diagnostic to watch. Once the virtuals are in, the rapidity logarithm must appear *linearly*: one ℓ_k multiplying $\mathcal{M} \otimes (\text{LO})$. If you still get ℓ_k^2 , the subtraction term is still transversely divergent and the $\log \frac{1}{\xi}$ from the z integral is still hitting the $\frac{1}{\xi}$ pole. That double logarithm is the symptom; the missing virtuals are the cause.

zint/src/zdiv_direct.py — the truncated z -integrals; zint/src/exactness.py — the exactness of the split at finite Λ . Output in zint/RESULTS_zdiv.txt.